A. Appendix

We provide the detailed descriptions for the experimental configurations of the control problems in the main text.

A.1. ANN structures

- For constructing the potential function V, we utilize the ICNN as (1):

$$\begin{aligned} \boldsymbol{z}_1 &= \sigma\left(W_0\boldsymbol{x}+\boldsymbol{b}_0\right), \\ \boldsymbol{z}_{i+1} &= \sigma\left(U_i\boldsymbol{z}_i+W_i\boldsymbol{x}+\boldsymbol{b}_i\right),\ i=1,\cdots,k-1, \\ p(\boldsymbol{x}) &\equiv \boldsymbol{z}_k,\ V(\boldsymbol{x})=\sigma\left[p(\boldsymbol{x})-p(\boldsymbol{0})\right]+\varepsilon\|\boldsymbol{x}\|^2, \end{aligned}$$

  where σ is the smoothed **ReLU** function as defined in the main text, $W_i \in \mathbb{R}^{h_i\times d}$, $U_i \in (\mathbb{R}_+\cup\{0\})^{h_i\times h_{i-1}}$, $\boldsymbol{x}\in\mathbb{R}^d$, and, for simplicity, this ICNN function is denoted by ICNN$(h_0,h_1,\cdots,h_{k-1})$;

- The class-$\mathcal{K}$ function α is constructed as:

$$\begin{aligned} \boldsymbol{q}_{i+1} &= \mathrm{ReLU}\left(W_i\boldsymbol{q}_i+\boldsymbol{b}_i\right), i=0,\cdots,k-2, \\ \boldsymbol{q}_k &= \mathrm{ELU}\left(W_{k-1}\boldsymbol{q}_{k-1}+b_{k-1}\right),\ \alpha(x)=\int_0^x q_k(s)\,\mathrm{d}s, \end{aligned}$$

  where $W_i \in \mathbb{R}^{h_{i+1}\times h_i}$, and this class-$\mathcal{K}$ function is denoted by $\mathcal{K}(h_0,h_1,\cdots,h_k)$;

- The neural control function (nonlinear version) is constructed as:

$$\begin{aligned} \boldsymbol{z}_{i+1} &= \mathcal{F}\left(W_i\boldsymbol{z}_i+\boldsymbol{b}_i\right),\ i=0,\cdots,k-1, \\ \mathbf{NN}(\boldsymbol{x}) &\equiv W_k\boldsymbol{z}_k,\ \boldsymbol{u}(\boldsymbol{x})=\mathrm{diag}(\boldsymbol{x})\,\mathbf{NN}(\boldsymbol{x}), \end{aligned}$$

  where $\mathcal{F}(\cdot)$ is the activation function, $W_i\in\mathbb{R}^{h_{i+1}\times h_i}$, this controller is denoted by Control$(h_0,h_1,\cdots,h_{k+1})$.

A.2. Double Pendulum

Here we model the state of the fully actuated noise-perturbed double pendulum as $\dot{\theta}_{1,2}=z_{1,2}+g_{1,2}u_{1,2}$,
$\mathrm{d}z_1=(m_2g\sin\theta_2\cos(\theta_1-\theta_2)-m_2\sin(\theta_1-\theta_2)[l_1z_1^2\cos(\theta_1-\theta_2)+l_2z_2^2]-(m_1+m_2)g\sin\theta_1)/(l_1m_1+m_2\sin^2(\theta_1\theta_2)+g_3u_3)\mathrm{d}t+\sin(\theta_1)\mathrm{d}W_1(t)$,
$\mathrm{d}z_2=[((m_1+m_2)[l_1z_1^2\sin(\theta_1-\theta_2)-g\sin\theta_2+g\sin\theta_1\cos(\theta_1-\theta_2)]+m_2l_2z_2^2\sin(\theta_1-\theta_2)\cos(\theta_1-\theta_2))/(l_2[m_1+m_2\sin^2(\theta_1-\theta_2)])+g_4u_4]\mathrm{d}t+\sin(\theta_2)\mathrm{d}W_2(t)$.

We transform the equilibrium to the zero by the coordinate transformation $\tilde{\theta}_{1,2}=\theta_{1,2}-\pi$. The safe region $h(\theta_{1,2},z_{1,2})=\sin(\theta_1)+0.5 \iff h(\tilde{\theta}_{1,2},z_{1,2})=0.5-\sin(\tilde{\theta}_1)$. Then we consider the dynamic for $\boldsymbol{x}=(\tilde{\theta}_{1,2},z_{1,2})$, and the control signals $\boldsymbol{u}(\boldsymbol{x})=(u_1(\boldsymbol{x}),u_2(\boldsymbol{x}),u_3(\boldsymbol{x}),u_4(\boldsymbol{x}))$ directly act on $\boldsymbol{x}$ with $\boldsymbol{g}=\mathrm{diag}(g_1,g_2,g_3,g_4)=\mathrm{diag}(0,0,1,1)$. For training controller $\boldsymbol{u}$, we sample 500 data from the safe region $[-7\pi/6,\pi/6]\times[-5,5]^3$, we construct the ANNs as follows.

ESS-PPO We parameterize $V(\boldsymbol{x})$ as ICNN(4, 12, 12, 1), $\alpha(x)$ as $\mathcal{K}(1,10,10,1)$, $\boldsymbol{u}(\boldsymbol{x})$ as Control(4, 12, 12, 4). We set $\varepsilon=$ 1e-3, $c=-0.1$. We train the parameters with lr $=0.1$ for 300 steps.

GP-MPC Since we have known the dynamics, there is no need to fitting the dynamics as GP flows from data. We obtain the dynamics of moment $\boldsymbol{z}=(\mu,\Sigma)$ of distribution $p(\boldsymbol{x})$ as

$$\begin{aligned} \mu_{t+1} &= \mu_t+dt\left(\cdot f\left(\mu_t\right)+\boldsymbol{u}_t\right), \\ \Sigma_{t+1} &= \Sigma_t+dt\cdot g\left(\mu_t\right), \end{aligned}$$

where f, g are the drift and diffusion terms, respectively. We denote the above dynamic as

$$\boldsymbol{z}_{t+1}=\tilde{f}\left(\boldsymbol{z}_t,\boldsymbol{u}_t\right).$$

Then we solve the MPC problem as

$$\begin{aligned} \min_{\boldsymbol{u}_{0:k-1}}\ &\Phi\left(\boldsymbol{z}_k\right)+\sum_{i=0}^{k-1}l\left(\boldsymbol{z}_i,u_i\right), \\ &\text{s.t. } \boldsymbol{z}_{i+1}=\tilde{f}\left(\boldsymbol{z}_i,\boldsymbol{u}_i\right),\ \boldsymbol{z}_0=(\boldsymbol{x},\boldsymbol{0}), \end{aligned}$$

and apply the first control element $\boldsymbol{u}_0$ to the current dynamic. We use the PMP method as proposed in (2) to solve the above problem. For safety and stability, we select cost function as

$$\begin{aligned} \Phi(\boldsymbol{z}) &= p_1\|\mu\|^2+p_3\|\Sigma\|^2, \\ l(\boldsymbol{z},\boldsymbol{u}) &= p_2\mathrm{ReLU}\left[\sin\left(\tilde{\theta}_1\right)-0.5\right]+\|\boldsymbol{z}\|^2. \end{aligned}$$

We set $p_1=6, p_2=2.5, p_3=0.5, k=10$ and find the optimal control sequence via the gradient descent of the Hamiltonian function in PMP.

QP The object function in QP is set as

$$\begin{aligned} \min_{\boldsymbol{u},d_1,d_2}\ &\frac{1}{2}\|\boldsymbol{u}\|^2+p_1d_1^2+p_2d_2^2, \\ &\text{s.t.}\mathcal{L}_{\boldsymbol{u}}V-\varepsilon V\le d_1, \\ &\mathcal{L}_{\boldsymbol{u}}\left[\frac{1}{h(\boldsymbol{x})}\right]-\gamma\frac{1}{h(\boldsymbol{x})}\le d_2, \end{aligned}$$

where $d_{1,2}$ are the relaxation numbers. We choose $V=\frac{1}{2}\|\boldsymbol{x}\|^2, p_1=10, p_2=10, \varepsilon=0.5, \gamma=2$.

We use Euler-Maruyama numerical scheme to simulate the system without and with control, and the random seeds are set as $\{1,4,6,8,9,14,15,16,17,18\}$.

A.3. Planar kinematic bicycle model

Here we model the state of the common noise-perturbed kinematic bicycle system as $\boldsymbol{x}=(x,y,\theta,v)^\top$, where x, y are the coordinate positions in phase plane, θ is the heading, v is the velocity. The dynamic is as follows,

$$\begin{aligned} \mathrm{d}x(t) &= \left[v(t)\cos\theta(t)+u_1\right]\mathrm{d}t+x(t)\,\mathrm{d}W_1(t) \\ \mathrm{d}y(t) &= \left[v(t)\sin\theta(t)+u_2\right]\mathrm{d}t+y(t)\,\mathrm{d}W_2(t) \\ \mathrm{d}\theta(t) &= \left[v(t)+u_3\right]\mathrm{d}t \\ \mathrm{d}v(t) &= \left[x(t)^2+y(t)^2+u_4\right]\mathrm{d}t \end{aligned}$$

The safe function is $h(x,y,\theta,v)=2^2-(x^2+y^2)$. For training controller $\boldsymbol{u}$, we sample 500 data $(r\cos(w), r\sin(w), \theta, v)$

from the safe region $(r, w, \theta, v) \in [0,3] \times [0, 2\pi] \times [-3,3]^2$, we construct the NNs as follows.

ESS-PPO We parameterize $V(\boldsymbol{x})$ as ICNN(4, 12, 12, 1), $\alpha(x)$ as $\mathcal{K}(1, 10, 10, 1)$, $\boldsymbol{u}(\boldsymbol{x})$ as Control(4, 12, 12, 4). We set ε = 1e-3, $c = -0.5$. We train the parameters with lr = 0.05 for 500 steps.

GP-MPC Since we have known the dynamics, there is no need to fitting the dynamics as GP flows from data. We obtain the dynamics of moment $\boldsymbol{z} = (\mu, \Sigma)$ of distribution $p(\boldsymbol{x})$ as $\mu_{t+1} = \mu_t + dt(\cdot f(\mu_t) + \boldsymbol{u}_t)$, $\Sigma_{t+1} = \Sigma_t + dt \cdot g(\mu_t)$, where f, g are the drift and diffusion terms, respectively. We denote the above dynamic as $\boldsymbol{z}_{t+1} = \tilde{f}(\boldsymbol{z}_t, \boldsymbol{u}_t)$. Then we solve the MPC problem as

$$\min_{\boldsymbol{u}_{0:k-1}} \Phi\left(\boldsymbol{z}_k\right) + \sum_{i=0}^{k-1} l\left(\boldsymbol{z}_i, u_i\right),$$
$$\text{s.t. } \boldsymbol{z}_{i+1} = \tilde{f}\left(\boldsymbol{z}_i, \boldsymbol{u}_i\right),\ \boldsymbol{z}_0 = (\boldsymbol{x}, \boldsymbol{0}),$$

and apply the first control element $\boldsymbol{u}_0$ to the current dynamic. We use the PMP method as proposed in (2) to solve the above problem. For safety and stability, we select cost function as

$$\Phi(\boldsymbol{z}) = p_1\|\mu\|^2 + p_3\|\Sigma\|^2,$$
$$l(\boldsymbol{z}, \boldsymbol{u}) = p_2 \text{ReLU}\left(\boldsymbol{z}_x^2 + \boldsymbol{z}_y^2 - 2^2\right) + \|\boldsymbol{z}\|^2.$$

We set $p_1 = 5, p_2 = 2.5, p_3 = 0.5, k = 5$ and find the optimal control sequence via the gradient descent of the Hamiltonian function in PMP.

QP The object function in QP is set as

$$\min_{\boldsymbol{u}, d_1, d_2} \frac{1}{2}\|\boldsymbol{u}\|^2 + p_1 d_1^2 + p_2 d_2^2,$$
$$\text{s.t.} \mathcal{L}_{\boldsymbol{u}} V - \varepsilon V \le d_1, \mathcal{L}_{\boldsymbol{u}}\left[\frac{1}{h(\boldsymbol{x})}\right] - \gamma\frac{1}{h(\boldsymbol{x})} \le d_2,$$

where $d_{1,2}$ are the relaxation numbers. We choose $V = \frac{1}{2}\|\boldsymbol{x}\|^2, p_1 = 10, p_2 = 10, \varepsilon = 0.5, \gamma = 2$.

We use Euler-Maruyama numerical scheme to simulate the system without and with control, and the random seeds are set as $\{0, 5, 7, 8, 10, 12, 14, 17, 18, 21\}$.

A.4. Fitzhugh-Nagumo model

Consider the coupled FHN model $\boldsymbol{x}_{1:50}$ as $\mathrm{d}\boldsymbol{x}_i = (f(\boldsymbol{x}_i) + u_i)\mathrm{d}t + \sum_{j=1}^{N} g(\boldsymbol{x}_j)\mathrm{d}W_t$ where $\boldsymbol{x}_i = v_i, w_i$, $f(\boldsymbol{x}_i) = (v_i - \frac{v_i^3}{3} - w_i + 1, 0.1(v_i + 0.7 - 0.8w_i))^\top$, $g(\boldsymbol{x}_i) = (\frac{1}{3}v_i, 0)^\top$. Let $\boldsymbol{\delta} = (\tilde{\boldsymbol{x}}_1, \cdots, \tilde{\boldsymbol{x}}_{50})^\top = (\boldsymbol{x}_1 - \boldsymbol{s}, \cdots, \boldsymbol{x}_{50} - \boldsymbol{s})^\top$, where $\boldsymbol{s}$ is on the synchronization manifold s.t. $\dot{\boldsymbol{s}} = f(\boldsymbol{s})$, then we have the variance equation $\mathrm{d}\boldsymbol{\delta} = (I \otimes \nabla f)\boldsymbol{\delta}\mathrm{d}t + (I \otimes \nabla g)\boldsymbol{\delta}\mathrm{d}W_t$. We only need to stabilize the variance $\boldsymbol{\delta}$ to zero, and the safe function is set as $h(\boldsymbol{\delta}) = 5^2 - \max_{1\le i\le 50}(\tilde{v}_i^2, \tilde{w}_i^2)$. We parameterize $V(\boldsymbol{\delta})$ as ICNN(100, 100, 100, 1), $\alpha(x)$ as $\mathcal{K}(1, 10, 10, 1)$, $\boldsymbol{u}(\boldsymbol{\delta})$ as Control(100, 200, 200, 100). In practice, we select mean field $1/N\sum_{j=1}^{N}\boldsymbol{x}_j$ as $\boldsymbol{s}$ and obtain controller $\boldsymbol{u}(\boldsymbol{\delta})$, which directly acts on $\boldsymbol{x}$ with $\boldsymbol{g} = \boldsymbol{I}_{100\times 100}$. We set ε = 1e-3, $c = -0.1$. We train the parameters with lr = 0.1 for 300 steps. We use Euler-Maruyama numerical scheme to simulate the system without and with control, and the random seeds are set as $\{1, 4, 5, 9, 15\}$.

A.5. Model in Example 3

We adopt the CBF $\hat{B}$ in Example 3 with relative degree 1, and we sample 5000 data from the safe region C_2 as defined in Example 3. We parameterize $V(\boldsymbol{x})$ as ICNN(2, 12, 12, 1), $\alpha(x)$ as $\mathcal{K}(1, 10, 10, 1)$, $\boldsymbol{u}(\boldsymbol{x})$ as Control(2, 6, 6, 2). We set ε = 1e-3, $c = -2$. We train the parameters with lr = 0.05 for 500 steps. We use Euler-Maruyama numerical scheme to simulate the system with ANN controller and LCk with $k = 1, \cdots, 10$, and the random seeds are set as $\{3, 5, 6, 7, 8\}$.

References

[1] B. Amos, L. Xu, and J. Z. Kolter, "Input convex neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, PMLR, pp. 146–155, 2017.

[2] S. Kamthe and M. P. Deisenroth, "Data-efficient reinforcement learning with probabilistic model predictive control," in *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics*, PMLR, pp. 1701–1710, 2018.